

15

A

total $E \propto \omega^2 x_0^2 \propto \left(\frac{1}{T}\right)^2 x_0^2$ where T is period and x_0 amplitude

$$\frac{E_2}{E_1} = \left(\frac{T_1}{T_2}\right)^2 \left[\frac{(x_0)_2}{(x_0)_1}\right]^2 = \left(\frac{T_1}{3T_1}\right)^2 \left[\frac{2(x_0)_1}{(x_0)_1}\right]^2 = \frac{4}{9} \Rightarrow E_2 = \frac{4}{9}E_1 = \frac{4}{9}(18 \text{ mJ}) = 8 \text{ mJ}$$

16

B